Microservice Chain

$r = < v_s, m_1, m_2, \dots, m_L, v_d >$

Computation queue length

ISL Capacity

CPU

Mem

Storage

$< \mathcal{G}^a, \mathbf{c}_{r,i}, \mathbf{x}_v, \mathbf{h}_v^l, \mathcal{M}, B, \dots >$

GNN-PPO Service-Satellite Selection Agent

1

Shortest Path, but heavy CPU queue

Idle CPU, but long transmission path

Optimal balance

2 Slot-aware Service Routing

Intermediate data

Time slot t_1

Traffic Splitting

ISLs in polar region

Time slot t_2

t_1 t_2 ⋯

Stage ①

T_r^{e2e}: End-to-end latency

E_r^{tot}: Execution energy

Service request tracing data collector

3 Bandit-based Redeployment

Y_{dep} slot windows

Hotspot Services

$k_1 = (o_1, m_1, p_s^1, p_t^1)$

$k_2 = (o_2, m_2, p_s^2, p_t^2)$

$k_3 = (o_3, m_3, p_s^3, p_t^3)$

Bandit arms

Add

Move

Remove

p_1

p_2

Apply actions

Reward vector